

16

Thermodynamic potentials

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The internal energy U of a system is a function of state, which means that a system undergoes the same change in U when we move it from one equilibrium state to another, irrespective of which route we take through parameter space. This makes U a very useful quantity, though not a uniquely useful quantity. In fact, we can make a number of other functions of state, simply by adding to U various other combinations of the functions of state p , V , T and S in such a way as to give the resulting quantity the dimensions of energy. These new functions of state are called **thermodynamic potentials**, and examples include $U + TS$, $U - pV$, $U + 2pV - 3TS$. However, most thermodynamic potentials that one could pick are really not very useful (including the ones we've just used as examples!) but three of them are extremely useful and are given special symbols: $H = U + pV$, $F = U - TS$ and $G = U + pV - TS$. In this chapter, we will explore why these three quantities are so useful. First, however, we will review some properties concerning the internal energy U .

16.1 Internal energy, U

Let us review the results concerning the internal energy that were derived in Section 14.3. Changes in the internal energy U of a system are given by the first law of thermodynamics written in the form (eqn 14.17):

$$dU = TdS - pdV. \quad (16.1)$$

This equation shows that the *natural variables*¹ to describe U are S and V , since changes in U are due to changes in S and/or V . Hence we write $U = U(S, V)$ to show that U is a function of S and V . Moreover, if S and V are held constant for the system, then

$$dU = 0, \quad (16.2)$$

which is the same as saying that U is a constant. Equation 16.1 implies that the temperature T can be expressed as a differential of U using

$$T = \left(\frac{\partial U}{\partial S} \right)_V, \quad (16.3)$$

and similarly the pressure p can be expressed as

$$p = - \left(\frac{\partial U}{\partial V} \right)_S. \quad (16.4)$$

¹See Section 14.3.

We also have that for **isochoric** processes (where isochoric means that V is constant),

$$dU = TdS, \quad (16.5)$$

and for reversible² isochoric processes

$$dU = dQ_{\text{rev}} = C_V dT, \quad (16.6)$$

and hence

$$\Delta U = \int_{T_1}^{T_2} C_V dT. \quad (16.7)$$

This is only true for systems held at constant volume; we would like to be able to extend this to systems held at constant pressure (an easier constraint to apply experimentally), and this can be achieved using the thermodynamic potential called enthalpy which we describe next.

16.2 Enthalpy, H

We define the **enthalpy** H by

$$\boxed{H = U + PV}. \quad (16.8)$$

This definition together with eqn 16.1 imply that

$$\begin{aligned} dH &= TdS - pdV + pdV + Vdp \\ &= TdS + Vdp. \end{aligned} \quad (16.9)$$

The natural variables for H are thus S and p , and we have that $H = H(S, p)$. We can therefore immediately write down that for a **isobaric** (i.e. constant pressure) process,

$$dH = TdS, \quad (16.10)$$

and for a reversible isobaric process

$$dH = dQ_{\text{rev}} = C_p dT, \quad (16.11)$$

so that

$$\Delta H = \int_{T_1}^{T_2} C_p dT. \quad (16.12)$$

This shows the importance of H , that for reversible isobaric processes the enthalpy represents the heat absorbed by the system.³ Isobaric conditions are relatively easy to obtain: an experiment which is open to the air in a laboratory is usually at constant pressure since pressure is provided by the atmosphere.⁴ We also conclude from eqn 16.9 that if both S and p are constant, we have that $dH = 0$.

Equation 16.9 also implies that

$$T = \left(\frac{\partial H}{\partial S} \right)_p, \quad (16.13)$$

²For a reversible process, $dQ = T dS$, see Section 14.3.

³If you add heat to the system at constant pressure, the enthalpy H of the system goes up. If heat is provided by the system to its surroundings H goes down.

⁴At a given latitude, the atmosphere provides a constant pressure, small changes due to weather fronts notwithstanding.

and

$$V = \left(\frac{\partial H}{\partial p} \right)_S. \quad (16.14)$$

Both U and H suffer from the drawback that one of their natural variables is the entropy S , which is not a very easy parameter to vary in a lab. It would be more convenient if we could substitute that for the temperature T , which is, of course, a much easier quantity to control and to vary. This is accomplished for both of our next two functions of state, the Helmholtz and Gibbs functions.

16.3 Helmholtz function, F

⁵This is sometimes called Helmholtz free energy.

We define the **Helmholtz function**⁵ using

$$\boxed{F = U - TS}. \quad (16.15)$$

Hence we find that

$$\begin{aligned} dF &= TdS - pdV - TdS - SdT \\ &= -SdT - pdV. \end{aligned} \quad (16.16)$$

This implies that the natural variables for F are V and T , and we can therefore write $F = F(T, V)$. For an *isothermal* process (constant T), we can simplify eqn 16.16 further and write that

$$dF = -pdV, \quad (16.17)$$

and hence

$$\Delta F = - \int_{V_1}^{V_2} p dV. \quad (16.18)$$

Hence a positive change in F represents reversible work done on the system by the surroundings, while a negative change in F represents reversible work done on the surroundings by the system. As we shall see in Section 16.5, F represents the maximum amount of work you can get out of a system at constant temperature, since the system will do work on its surroundings until its Helmholtz function reaches a minimum. Equation 16.16 implies that the entropy S can be written as

$$S = - \left(\frac{\partial F}{\partial T} \right)_V, \quad (16.19)$$

and the pressure p as

$$p = - \left(\frac{\partial F}{\partial V} \right)_T. \quad (16.20)$$

If T and V are constant, we have that $dF = 0$ and F is a constant.

16.4 Gibbs function, G .

We define the **Gibbs function**⁶ using

$$G = H - TS. \quad (16.21)$$

Hence we find that

$$\begin{aligned} dG &= TdS + Vdp - TdS - SdT \\ &= -SdT + Vdp, \end{aligned} \quad (16.22)$$

and hence the natural variables of G are T and p . [Hence we can write $G = G(T, p)$.]

Having T and p as natural variables is particularly convenient as T and p are the easiest quantities to manipulate and control for most experimental systems. In particular, note that if T and p are constant, $dG = 0$. Hence G is conserved in any isothermal isobaric process.⁷

The expression in eqn 16.22 allows us to write down expressions for entropy and volume as follows:

$$S = - \left(\frac{\partial G}{\partial T} \right)_p \quad (16.23)$$

and

$$V = - \left(\frac{\partial G}{\partial p} \right)_T. \quad (16.24)$$

We have now defined the four main thermodynamic potentials which are useful in much of thermal physics: the internal energy U , the enthalpy H , the Helmholtz function F and the Gibbs function G . Before proceeding further, we summarize the main equations which we have used so far.

⁶This is sometimes called the Gibbs free energy.

⁷For example, at a phase transition between two different phases (call them phase 1 and phase 2), there is phase co-existence between the two phases at the same pressure at the transition temperature. Hence the Gibbs functions for phase 1 and phase 2 must be equal at the phase transition. This will be particularly useful for us in Chapter 28.

Function of state		Differential	Natural variables	First derivatives	
Internal energy	U	$dU = TdS - pdV$	$U = U(S, V)$	$T = \left(\frac{\partial U}{\partial S} \right)_V,$	$p = - \left(\frac{\partial U}{\partial V} \right)_S$
Enthalpy	$H = U + pV$	$dH = TdS + Vdp$	$H = H(S, p)$	$T = \left(\frac{\partial H}{\partial S} \right)_p,$	$V = \left(\frac{\partial H}{\partial p} \right)_S$
Helmholtz function	$F = U - TS$	$dF = -SdT - pdV$	$F = F(T, V)$	$S = - \left(\frac{\partial F}{\partial T} \right)_V,$	$p = - \left(\frac{\partial F}{\partial V} \right)_T$
Gibbs function	$G = H - TS$	$dG = -SdT + Vdp$	$G = G(T, p)$	$S = - \left(\frac{\partial G}{\partial T} \right)_p,$	$V = \left(\frac{\partial G}{\partial p} \right)_T$

Note that to derive these equations quickly, all you need to do is memorize the definitions of H , F and G and the first law in the form $dU = TdS - pdV$ and the rest can be written down straightforwardly.

Example 16.1

Show that $U = -T^2 \left(\frac{\partial}{\partial T} \right)_V \frac{F}{T}$ and $H = -T^2 \left(\frac{\partial}{\partial T} \right)_p \frac{G}{T}$.

Solution:

Using the expressions

$$S = - \left(\frac{\partial F}{\partial T} \right)_V, \quad \text{and} \quad S = - \left(\frac{\partial G}{\partial T} \right)_p,$$

we can write down

$$U = F + TS = F - T \left(\frac{\partial F}{\partial T} \right)_V = -T^2 \left(\frac{\partial(F/T)}{\partial T} \right)_V, \quad (16.25)$$

and

$$H = G + TS = G - T \left(\frac{\partial G}{\partial T} \right)_p = -T^2 \left(\frac{\partial(G/T)}{\partial T} \right)_p. \quad (16.26)$$

These equations are known as the **Gibbs–Helmholtz equations** and are useful in chemical thermodynamics.

16.5 Availability

We want to try now to work out how to find the equilibrium properties of a system when it is placed in contact with its surroundings. In general, a system is able to exchange heat with its surroundings and also to do work on its surroundings. Let us now consider a system in contact with surroundings which are at temperature T_0 and pressure p_0 (see Fig. 16.1). Let us consider what happens when we transfer energy dU and volume dV from the surroundings, to the system. The internal energy of the surroundings changes by dU_0 , where

$$dU_0 = -dU = T_0 dS_0 - p_0(-dV), \quad (16.27)$$

where the minus signs express the fact that the energy and volume in the surroundings are *decreasing*. We can rearrange this expression to give the change of entropy in the surroundings as

$$dS_0 = - \left[\frac{dU + p_0 dV}{T_0} \right]. \quad (16.28)$$

If the entropy of the system changes by dS , then the total change of entropy dS_{tot} is

$$dS_{\text{tot}} = dS_0 + dS, \quad (16.29)$$

and the second law of thermodynamics implies that $dS_{\text{tot}} \geq 0$. Using eqns 16.28 and 16.29, we have that

$$T_0 dS_{\text{tot}} = -[dU + p_0 dV - T_0 dS] \geq 0. \quad (16.30)$$

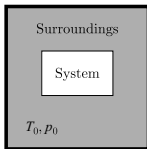


Fig. 16.1 A system in contact with surroundings at temperature T_0 and pressure p_0 .

Hence

$$dU + p_0 dV - T_0 dS \leq 0. \quad (16.31)$$

We now define the **availability** A by

$$A = U + p_0 V - T_0 S, \quad (16.32)$$

and because p_0 and T_0 are constants, then

$$dA = dU + p_0 dV - T_0 dS. \quad (16.33)$$

Hence eqn 16.31 becomes

$$\boxed{dA \leq 0.} \quad (16.34)$$

We have derived this inequality from the second law of thermodynamics. It demonstrates that changes in A are always negative. As a system settles down to equilibrium, any changes will always force A downwards. Once the system has reached equilibrium, A will be constant at this minimum level. Hence equilibrium can only be achieved by minimizing A . However, the type of equilibrium achieved depends on the nature of the constraints, as we will now show.

- *System with fixed entropy and volume:*

In this case $dS = dV = 0$ and hence eqns 16.33 and 16.34 imply

$$dA = dU \leq 0, \quad (16.35)$$

so we must *minimize* U to find the equilibrium state of this system.

- *System with fixed entropy and pressure:*

In this case $dS = dp = 0$, and hence eqn 16.33 implies

$$dA = dU + p_0 dV. \quad (16.36)$$

The change in enthalpy is $dH = dU + p dV + V dp$, and since $p = p_0$ and $dp = 0$, we have that

$$dH = dU + p_0 dV, \quad (16.37)$$

and hence

$$dA = dH \leq 0, \quad (16.38)$$

so we must *minimize* H to find the equilibrium state.

- *System thermally isolated and with fixed volume:*

Since no heat can enter the system and the system can do no work on its surroundings, $dU = 0$. Hence eqn 16.33 becomes $dA = -T_0 dS$ and hence $dA \leq 0$ implies that $dS \geq 0$. Thus we must *maximise* S to find the equilibrium state.

- *System with fixed volume at constant temperature:*

$dA = dU - T_0 dS \leq 0$, but because $dT = 0$ and $dF = dU - T_0 dS - SdT = dU - T_0 dS$, we have that

$$dA = dF \leq 0, \quad (16.39)$$

so we must *minimize* F to find the equilibrium state.

- System at constant pressure and temperature:

Eqn 16.33 gives $dA = dU - T_0 dS + p_0 dV \leq 0$. We can write dG (from the definition $G = H - TS$) as

$$dG = dU + p_0 dV + V dp - T_0 dS - S dT = dU - T_0 dS + p_0 dV, \quad (16.40)$$

since $dp = dT = 0$, and hence

$$dA = dG \leq 0, \quad (16.41)$$

so we must *minimize* G to find the equilibrium state.

Example 16.2

Chemistry laboratories are usually at constant pressure. If a chemical reaction is carried out at constant pressure, then by eqn 16.10 we have that

$$\Delta H = \Delta Q, \quad (16.42)$$

and hence ΔH is the reversible heat added to the system, i.e. the heat absorbed by the reaction. (Recall that our convention is that ΔQ is the heat *entering* the system, and in this case the system is the reacting chemicals.)

- If $\Delta H < 0$, the reaction is called **exothermic** and heat will be emitted.
- If $\Delta H > 0$, the reaction is called **endothermic** and heat will be absorbed.

However, this does not tell you whether or not a chemical reaction will actually proceed. Usually reactions occur⁸ at constant T and p , so if the system is trying to minimize its availability, then we need to consider ΔG . The second law of thermodynamics (via eqn 16.34 and hence eqn 16.41) therefore implies that a chemical system will minimize G , so that if $\Delta G < 0$, the reaction may spontaneously occur.⁹

⁸The temperature may rise *during* a reaction, but if the final products cool to the original temperature, one only needs to think about the beginning and end points, since G is a function of state.

⁹However, one may also need to consider the kinetics of the reaction. Often a reaction has to pass via a metastable intermediate state which may have a higher Gibbs function, so the system cannot spontaneously lower its Gibbs function without having it slightly raised first. This gives a reaction an **activation energy** which must be added before the reaction can proceed, even though the completion of the reaction gives you all that energy back and more.

16.6 Maxwell's relations

In this section, we are going to derive four equations which are known as Maxwell's relations. These equations are very useful in solving problems in thermodynamics, since each one relates a partial differential between quantities that can be hard to measure to a partial differential between quantities that can be much easier to measure. The derivation proceeds along the following lines: a state function f is a function of variables x and y . A change in f can be written as

$$df = \left(\frac{\partial f}{\partial x} \right)_y dx + \left(\frac{\partial f}{\partial y} \right)_x dy. \quad (16.43)$$

Because df is an exact differential (see Appendix C.7), we have that

$$\left(\frac{\partial^2 f}{\partial x \partial y}\right) = \left(\frac{\partial^2 f}{\partial y \partial x}\right). \quad (16.44)$$

Hence writing

$$F_x = \left(\frac{\partial f}{\partial x}\right)_y \quad \text{and} \quad F_y = \left(\frac{\partial f}{\partial y}\right)_x, \quad (16.45)$$

we have that

$$\left(\frac{\partial F_y}{\partial x}\right) = \left(\frac{\partial F_x}{\partial y}\right). \quad (16.46)$$

We can now apply this idea to each of the state variables U , H , F and G in turn.

Example 16.3

The Maxwell relation based on G can be derived as follows. We write down an expression for dG :

$$dG = -SdT + Vdp. \quad (16.47)$$

We can also write

$$dG = \left(\frac{\partial G}{\partial T}\right)_p dT + \left(\frac{\partial G}{\partial p}\right)_T dp, \quad (16.48)$$

and hence we can write $S = -\left(\frac{\partial G}{\partial T}\right)_p$ and $V = \left(\frac{\partial G}{\partial p}\right)_T$. Because dG is an exact differential, we have that

$$\left(\frac{\partial^2 G}{\partial T \partial p}\right) = \left(\frac{\partial^2 G}{\partial p \partial T}\right), \quad (16.49)$$

and hence we have the following Maxwell relation:

$$-\left(\frac{\partial S}{\partial p}\right)_T = \left(\frac{\partial V}{\partial T}\right)_p \quad (16.50)$$

This reasoning can be applied to each of the thermodynamic potentials U , H , F and G to yield the four **Maxwell relations**:

Maxwell's relations:

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V \quad (16.51)$$

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p \quad (16.52)$$

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V \quad (16.53)$$

$$\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p \quad (16.54)$$

¹⁰If you do, however, insist on memorizing them, then lots of mnemonics exist. One useful way of remembering them is as follows. Each Maxwell relation is of the form

$$\left(\frac{\partial *}{\partial \dagger}\right)_* = \pm \left(\frac{\partial \dagger}{\partial *}\right)_{\dagger}$$

where the pairs of symbols which are similar to each other (* and *, or † and †) signify **conjugate variables**, so that their product has the dimensions of energy: e.g. T and S , and p and V . Thus you can notice that, for each Maxwell relation, terms diagonally opposite each other are conjugate variables. The variable held constant is conjugate to the one on the top of the partial differential. Another point is that you always have a minus sign when V and T are on the same side of equation.

These equations should not be memorized;¹⁰ rather it is better to remember how to derive them!

A more sophisticated way of deriving these equations based on Jacobians (which may not be everybody's taste) is outlined in the box below. It has the attractive virtue of producing all four Maxwell relations in one go by directly relating the work done and heat absorbed in a cyclic process, but the unfortunate vice of requiring easy familiarity with the use of Jacobian transformations.

An alternative derivation of Maxwell's relations:

The following derivation is more elegant, but requires a knowledge of Jacobians (see Appendix C.9): Consider a cyclic process which can be described in both the T - S and p - V planes. The internal energy U is a state function and therefore doesn't change in a cycle, so $\oint p dV = \oint T dS$, and hence we have

$$\oint \int dp dV = \oint \int dT dS, \quad (16.55)$$

so that the work done (the area enclosed by the cycle in the p - V plane) is equal to the heat absorbed (the area enclosed by the cycle in the T - S plane). However, one can also write

$$\oint \int dp dV \frac{\partial(T, S)}{\partial(p, V)} = \oint \int dT dS, \quad (16.56)$$

where $\partial(T, S)/\partial(p, V)$ is the Jacobian of the transformation from the p - V plane to the T - S plane, and so this implies that

$$\boxed{\frac{\partial(T, S)}{\partial(p, V)} = 1.} \quad (16.57)$$

This equation is sufficient to generate all four Maxwell relations via

$$\frac{\partial(T, S)}{\partial(x, y)} = \frac{\partial(p, V)}{\partial(x, y)}, \quad (16.58)$$

where (x, y) are taken as (i) (T, p) , (ii) (T, V) , (iii) (p, S) and (iv) (S, V) , and using the identities in Appendix C.9.

We will now give several examples of how Maxwell's relations can be used to solve problems in thermodynamics.

Example 16.4

Find expressions for $(\partial C_V/\partial p)_T$ and $(\partial C_V/\partial V)_T$ in terms of p , V and T .

Solution:

By the definitions of C_V and C_p we have that

$$C_V = \left(\frac{\partial Q}{\partial T}\right)_V = T \left(\frac{\partial S}{\partial T}\right)_V \quad (16.59)$$

and

$$C_p = \left(\frac{\partial Q}{\partial T}\right)_p = T \left(\frac{\partial S}{\partial T}\right)_p. \quad (16.60)$$

Now

$$\begin{aligned} \left(\frac{\partial C_p}{\partial p}\right)_T &= \left(\frac{\partial}{\partial p} T \left(\frac{\partial S}{\partial T}\right)_p\right)_T \\ &= T \left(\frac{\partial}{\partial p} \left(\frac{\partial S}{\partial T}\right)_p\right)_T \\ &= T \left(\frac{\partial}{\partial T} \left(\frac{\partial S}{\partial p}\right)_T\right)_p \end{aligned} \quad (16.61)$$

and using a Maxwell relation

$$\left(\frac{\partial C_p}{\partial p}\right)_T = -T \left(\frac{\partial}{\partial T} \left(\frac{\partial V}{\partial T}\right)_p\right)_p = -T \left(\frac{\partial^2 V}{\partial T^2}\right)_p. \quad (16.62)$$

Similarly

$$\left(\frac{\partial C_V}{\partial V}\right)_T = T \left(\frac{\partial^2 p}{\partial T^2}\right)_V. \quad (16.63)$$

Both the expressions in eqns 16.62 and 16.63 are zero for a perfect gas.

Before proceeding further with the examples, we will pause to list the tools which you have at your disposal to solve these sorts of problems. Any given problem may not require you to use all of these, but you may have to use more than one of these 'techniques'.

- (1) **Write down a function of state in terms of particular variables.**

If f is a function of x and y , so that $f = f(x, y)$, you then have immediately that

$$df = \left(\frac{\partial f}{\partial x}\right)_y dx + \left(\frac{\partial f}{\partial y}\right)_x dy. \quad (16.64)$$

- (2) **Use Maxwell's relations to transform the partial differential you start with into a more convenient one.**

Use the Maxwell relations in eqns 16.51–16.54.

- (3) **Invert a Maxwell relation using the reciprocal theorem.**
The reciprocal theorem states that

$$\left(\frac{\partial x}{\partial z}\right)_y = \frac{1}{\left(\frac{\partial z}{\partial x}\right)_y}, \quad (16.65)$$

and this is proved in Appendix C.6 (see eqn C.41).

- (4) **Combine partial differentials using the reciprocity theorem.**

The reciprocity theorem states that

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1, \quad (16.66)$$

which is proved in Appendix C.6 (see eqn C.42). This can be combined with the reciprocal theorem to write that

$$\left(\frac{\partial x}{\partial y}\right)_z = - \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x, \quad (16.67)$$

which is a very useful identity.

- (5) **Identify a heat capacity.**

Some of the partial differentials which appear in Maxwell's relations relate to real, measurable properties. As we have seen in Example 16.4, both $\left(\frac{\partial S}{\partial T}\right)_V$ and $\left(\frac{\partial S}{\partial T}\right)_p$ can be related to heat capacities:

$$\frac{C_V}{T} = \left(\frac{\partial S}{\partial T}\right)_V \quad \text{and} \quad \frac{C_p}{T} = \left(\frac{\partial S}{\partial T}\right)_p \quad (16.68)$$

- (6) **Identify a “generalized susceptibility”.**

A **generalized susceptibility** quantifies how much a particular variable changes when a generalized force is applied. A **generalized force** is a variable such as T or p which is a differential of the internal energy with respect to some other parameter.¹¹ An example of a generalized susceptibility is $\left(\frac{\partial V}{\partial T}\right)_x$ which, you will recall, answers the question “keeping x constant, how much does the volume change when you change the temperature?”. It is related to the thermal expansivity at constant x , where x is pressure or entropy. Thus the **isobaric expansivity** β_p is defined as

$$\beta_p = \frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_p, \quad (16.69)$$

while the **adiabatic expansivity** β_S is defined as

$$\beta_S = \frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_S. \quad (16.70)$$

Expansivities measure the *fractional* change in volume with a change in temperature.

¹¹Recall that $T = (\partial U/\partial S)_V$ and $p = -(\partial U/\partial V)_S$.

Another useful generalized susceptibility is the compressibility. This quantifies how large a fractional volume change you achieve when you apply pressure. The **isothermal compressibility** κ_T is defined as

$$\kappa_T = -\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T, \quad (16.71)$$

while the **adiabatic compressibility** κ_S is defined as

$$\kappa_S = -\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_S. \quad (16.72)$$

Both quantities have a minus sign so that the compressibilities are positive (this is because things get smaller when you press them, so fractional volume changes are negative when positive pressure is applied). None of these expansivities or compressibilities appears directly in a Maxwell relation, but each can easily be related to those that do using the reciprocal and reciprocity theorems.

Example 16.5

By considering $S = S(T, V)$, show that $C_p - C_V = VT\beta_p^2/\kappa_T$.

Solution:

Considering $S = S(T, V)$ allows us to write down immediately that

$$dS = \left(\frac{\partial S}{\partial T} \right)_V dT + \left(\frac{\partial S}{\partial V} \right)_T dV. \quad (16.73)$$

Differentiating this equation with respect to T at constant p yields

$$\left(\frac{\partial S}{\partial T} \right)_p = \left(\frac{\partial S}{\partial T} \right)_V + \left(\frac{\partial S}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_p. \quad (16.74)$$

Now the first two terms can be replaced by C_p/T and C_V/T respectively, while use of a Maxwell relation and a partial differential identity (see eqn 16.67) yields

$$\left(\frac{\partial S}{\partial V} \right)_T = \left(\frac{\partial p}{\partial T} \right)_V = - \left(\frac{\partial p}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_p \quad (16.75)$$

and hence using eqns 16.69 and 16.71 we have that

$$C_p - C_V = \frac{VT\beta_p^2}{\kappa_T}. \quad (16.76)$$

The next example shows how to calculate the entropy of an ideal gas.

Example 16.6

Find the entropy of 1 mole of ideal gas.

Solution:

For one mole of ideal gas $pV = RT$. Consider the entropy S as a function of volume and temperature, i.e.

$$S = S(T, V), \quad (16.77)$$

so that

$$dS = \left(\frac{\partial S}{\partial T} \right)_V dT + \left(\frac{\partial S}{\partial V} \right)_T dV \quad (16.78)$$

$$= \frac{C_V}{T} dT + \left(\frac{\partial p}{\partial T} \right)_V dV, \quad (16.79)$$

using eqn 16.53 and eqn 16.68. The ideal gas law for 1 mole, $p = RT/V$, implies that

$$\left(\frac{\partial p}{\partial T} \right)_V = R/V, \quad (16.80)$$

and hence, if we integrate eqn 16.79,

$$S = \int \frac{C_V}{T} dT + \int \frac{RdV}{V}. \quad (16.81)$$

If C_V is not a function of temperature (which is true for an ideal gas) simple integration yields

$$S = C_V \ln T + R \ln V + \text{constant}. \quad (16.82)$$

The entropy of an ideal gas increases with increasing temperature and increasing volume.

The final example in this chapter shows how to prove that the ratio of the isothermal and adiabatic compressibilities, κ_T/κ_S , is equal to γ .

Example 16.7

Find the ratio of the isothermal and adiabatic compressibilities.

Solution:

This follows using straightforward manipulations of partial differentials. To begin with, we write

$$\frac{\kappa_T}{\kappa_S} = \frac{\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_T}{\frac{1}{V} \left(\frac{\partial V}{\partial p} \right)_S}, \quad (16.83)$$

which follows from the definition of κ_T and κ_S (eqns 16.71 and 16.72). Then we proceed as follows:

$$\begin{aligned}
 \frac{\kappa_T}{\kappa_S} &= \frac{-\left(\frac{\partial V}{\partial T}\right)_p \left(\frac{\partial T}{\partial p}\right)_V}{-\left(\frac{\partial V}{\partial S}\right)_p \left(\frac{\partial S}{\partial p}\right)_V} && \text{reciprocity theorem} \\
 &= \frac{\left(\frac{\partial V}{\partial T}\right)_p \left(\frac{\partial S}{\partial V}\right)_p}{\left(\frac{\partial p}{\partial T}\right)_V \left(\frac{\partial S}{\partial p}\right)_V} && \text{Maxwell's relations} \\
 &= \frac{\left(\frac{\partial S}{\partial T}\right)_p}{\left(\frac{\partial S}{\partial T}\right)_V} && \text{reciprocity theorem} \\
 &= \frac{C_p}{C_V} \\
 &= \gamma.
 \end{aligned} \tag{16.84}$$

We can show that this equation is correct for the case of an ideal gas as follows. Assuming the ideal gas equation $pV \propto T$, we have for constant temperature that

$$\frac{dp}{p} = -\frac{dV}{V}, \tag{16.85}$$

and hence

$$\kappa_T = \frac{1}{p}. \tag{16.86}$$

For an adiabatic change $p \propto V^{-\gamma}$ and hence

$$\frac{dp}{p} = -\gamma \frac{dV}{V}, \tag{16.87}$$

and hence

$$\kappa_S = \frac{1}{\gamma p}. \tag{16.88}$$

This agrees with eqn 16.84 above. We note that because κ_T is larger than κ_S (because $\gamma > 1$), the isotherms always have a smaller gradient than the adiabats on a p - V plot (see Fig. 12.1).

Chapter summary

- We define the following thermodynamic potentials:

$$U, \quad H = U + pV, \quad F = U - TS, \quad G = H - TS,$$

which are then related by the following differentials:

$$\begin{aligned} dU &= TdS - pdV \\ dH &= TdS + Vdp \\ dF &= -SdT - pdV \\ dG &= -SdT + Vdp \end{aligned}$$

- The availability A is given by $A = U + p_0V - T_0S$, and for any spontaneous change we have that $dA \leq 0$. This means that a system in contact with a reservoir (temperature T_0 , pressure p_0) will minimize A which means
 - minimizing U when S and V are fixed;
 - minimizing H when S and p are fixed;
 - minimizing F when T and V are fixed;
 - minimizing G when T and p are fixed.
- Four Maxwell relations can be derived from the boxed equations above, and used to solve many problems in thermodynamics.

Exercises

- (16.1) (a) Using the first law $dU = TdS - pdV$ to provide a reminder, write down the definitions of the four thermodynamic potentials U , H , F , G (in terms of U , S , T , p , V), and give dU , dH , dF , dG in terms of T , S , p , V and their derivatives.

- (b) Derive all the Maxwell relations.

- (16.2) (a) Derive the following general relations

$$(i) \quad \left(\frac{\partial T}{\partial V} \right)_U = -\frac{1}{C_V} \left[T \left(\frac{\partial p}{\partial T} \right)_V - p \right]$$

$$(ii) \quad \left(\frac{\partial T}{\partial V} \right)_S = -\frac{1}{C_V} T \left(\frac{\partial p}{\partial T} \right)_V$$

$$(iii) \quad \left(\frac{\partial T}{\partial p} \right)_H = \frac{1}{C_p} \left[T \left(\frac{\partial V}{\partial T} \right)_p - V \right]$$

In each case the quantity on the left hand side is the appropriate thing to consider for a particular type of expansion. State what type of expansion each refers to.

- (b) Using these relations, verify that for an ideal gas $(\partial T/\partial V)_U = 0$ and $(\partial T/\partial p)_H = 0$, and that $(\partial T/\partial V)_S$ leads to the familiar relation $pV^\gamma = \text{constant}$ along an isentrope.

- (16.3) Use the first law of thermodynamics to show that

$$\left(\frac{\partial U}{\partial V} \right)_T = \frac{C_p - C_V}{V\beta_p} - p, \quad (16.89)$$

where β_p is the coefficient of volume expansivity and the other symbols have their usual meanings.

- (16.4) (a) The natural variables for U are S and V . This

means that if you know S and V , you can find $U(S, V)$. Show that this also gives you simple expressions for T and p .

(b) Suppose instead that you know V , T and the function $U(T, V)$ (i.e. you have expressed U in terms of variables which are not all the natural variables of U). Show that this leads to a (much more complicated) expression for p , namely

$$\frac{p}{T} = \int \left(\frac{\partial U}{\partial V} \right)_T \frac{dT}{T^2} + f(V), \quad (16.90)$$

where $f(V)$ is some (unknown) function of V .

(16.5) Use thermodynamic arguments to obtain the general result that, for any gas at temperature T , the

pressure is given by

$$P = T \left(\frac{\partial P}{\partial T} \right)_V - \left(\frac{\partial U}{\partial V} \right)_T, \quad (16.91)$$

where U is the total energy of the gas.

(16.6) Show that another expression for the entropy per mole of an ideal gas is

$$S = C_p \ln T - R \ln p + \text{constant}. \quad (16.92)$$

(16.7) Show that the entropy of an ideal gas can be expressed as

$$S = C_V \ln \left(\frac{p}{\rho^\gamma} \right) + \text{constant}. \quad (16.93)$$